A pull between leftovers: why a square-counting graph energy shows none at a distance, and what a pull would need

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The author proposes that gravity is the wish of curved space to refold: that small curled leftovers of the change that made space pull on each other through the arrangement around them. We ask whether the family of graph energies obtained from combinatorial quantum gravity (CQG) by raising the coefficient λ of its local term above 1 can produce such a pull. At fixed wiring, exact results say it does not. The energy is a sum over links, so two defects that share no square are exactly additive; counting symmetries, in the cases computed, prefers symmetric placements, not near ones; and flat space is gapped in the extreme: every single move out of it costs 32, 64 or 128 units with four, six or eight links per point, at every λ. That such a gap allows only a short-range force is argued by analogy with the published literature, not derived for this model; a force with the wiring free to rearrange has not been measured, and with six links the relic such a force would act on has not been found. We then state what the published work says a pull needs (a mode with no gap, each relic a source for it in proportion to its energy, and a mode of the kind that makes like sources attract), and compute exactly what the simplest such mode a graph carries, a field on its points, would add: its fluctuations give a Casimir-type attraction under the field reading, a free-energy term falling roughly as r^{-4} in two directions and r^{-6} in three; with relics as its sources it gives a potential falling as $1/r$ in three directions, Newton's shape, carried by the emergent wiring; and the field barely changes which arrangement is favored. Such a field would put the pull in, not derive it. The author has decided to add a quantity on the points if it matches her account of gravity; one number per point is scalar gravity, which does not bend light, so that condition needs more than one (Sec. VI).

I. INTRODUCTION

The companion papers [1, 2] study a curled torus that opens into flat space, the energy it releases and the small curled leftover it leaves behind, the relic. The author's framework reads gravity as the refolding drive of curved space: relics, being curled, should draw one another together through the arrangement between them. That makes a definite question of the model: do two relics in flat space pull on each other, and if so, how does the pull fall with distance?

The energy is that of CQG [9] with the coefficient λ of its local term raised. On $2D$-regular bipartite graphs in which no two points share more than two neighbors (the hard-core rule),

$$\begin{aligned} H &= 16\Big(\tfrac{D(D-1)}{2}\,N - S\Big) + 4\lambda X, \\ X &= \textstyle\sum_e (S_e - s_0)_+, \end{aligned} \tag{1}$$

with S the number of 4-cycles (squares), S_e the number through link e, and $D(D-1)N/2$ the number of squares of flat space, where every link carries $s_0 = 2(D-1)$. Since $\sum_e S_e = 4S$ over the DN links and $a = a_+ - (-a)_+$, Eq. (1) is also

$$H = 4\sum_e \big[(s_0 - S_e)_+ + (\lambda - 1)(S_e - s_0)_+\big], \tag{2}$$

a sum over links that is ≥ 0 for $\lambda \geq 1$ and 0 on flat space. Only $\lambda = 1$ is CQG; everything run here is at $\lambda > 1$, next to it. The exact statements hold at every λ, including 1. The six- and eight-link energies are derived by us from the published definitions; the six-link kernel has not reproduced the published six-link curve, and there is no published eight-link curve, so every statement here at six and eight links is about our family and its kernel.

II. THREE EXACT FACTS

Additivity. By Eq. (2), H is a sum of terms on links, each depending only on the squares through that link. Two defects that share no square therefore cost exactly the sum of their separate costs, at every separation; an attraction appears only at contact (16 with four links, 64 with six), where they share squares. We checked it by construction for local defects at every separation on flat tori with four and with six links.

No counting pull at fixed wiring. With the points treated as interchangeable, as the author's framework treats them, each wiring counts once rather than once per labeling, so each placement of two identical defects is weighted, relative to named points, by its number of symmetries. With four links, for one simple defect on one torus, that number is one or two at ordinary separations and four only at exactly opposite placements; with six links it is 8 at generic separations, 16 exactly opposite, 32 along the defect's own axis at any separation and 4 on a diagonal. That is a preference for symmetric place-

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ments, not one that grows as the defects approach: not a pull.

A gapped vacuum. Flat space is an exact ground state for $\lambda \geq 1$ (Eq. 2), and the cheapest single move out of it costs 32, 64 and 128 with four, six and eight links, the same at every λ (found by trying every switch with four links; with six and eight, by trying switches between nearby links, a search checked against the full one on one six-link torus). At the couplings where curled arrangements are metastable ($g \lesssim 2$), a cold flat region accepts essentially no move at all; the relic itself anneals away within 500 to 80,500 sweeps at $g = 1.0$ to 1.5 and lasts only where its surroundings stay very cold [2]. With four links the relic is a local minimum (its energy is $24\lambda - 16$ and every single move out of it costs energy); with six links, a search at $\lambda = 1.02$ on the flat $6 \times 6 \times 6$ torus, over the 9,049 curled arrangements made by one or two switches within three steps of one point, testing the 30 lowest in energy, found no local minimum, so the object a pull would act on is not yet known to exist there.

III. WHY THE FACTS ARE ONE FACT

A force between two objects in a medium is carried by the medium's fluctuations, and its range is the range of their correlations [10]. A gapped system has correlations that decay exponentially [11]; a massive mediator gives Yukawa's e^{-mr}/r [12]. The model's author writes that both phases of the model are governed by a finite correlation length, which diverges only at the critical point [9]. Here "gapped" means that every single move out of flat space costs at least a fixed energy; the theorem of Ref. [11] is for quantum lattice Hamiltonians, and carrying it over to this classical ensemble, defined by its moves, is an analogy we have not proved. A vacuum whose every excitation costs 32 or more is the extreme case: at fixed wiring there is nothing to carry a force (the first two facts), and with the wiring free there is almost nothing that moves. So, by this argument, the absence of a pull between point defects is what this energy should give in its geometric phase (with the wiring free to rearrange, a force would be entropic; Sec. VI). One long-range force does exist in this family, of another kind: because two orders differ by a fixed energy per point, a boundary between them is pushed with a constant force, the force that drives the front of Ref. [1]. It acts between regions of different order, not between separated lumps, and it is not gravity.

We also measured, in the published model ($\lambda = 1$), that the curvature correlation falls to zero within two steps at every coupling and every size up to 676 points; the same measurement did not reproduce the model's published correlation length. That is consistent with a gap but does not show one: in Einstein's theory, which has a massless graviton, the curvature correlator is a contact term at tree level, and it is the volume correlator that carries the massless mode [13]. The gap itself is shown by the move costs of Sec. II.

IV. WHAT A PULL NEEDS

The published work names three things, all needed together. (i) A mode with no gap, so that the medium's correlations fall as a power, not an exponential. The robust classical route is a hard local constraint with many ground states, as in dimer models on bipartite lattices, where counting alone gives defects an inverse-square force [14, 15]; the quantum route is a gauge structure [16]. (ii) Each relic a source for that mode in proportion to its energy, a charge whose flux through every surrounding surface is the same; without it a gapless medium gives only the rapidly decaying Casimir-type forces that stiff inclusions feel in a membrane [10, 17]. (iii) A mode of the kind that makes like sources attract: a scalar, as in Nordström's theory [18], or the tensor of Einstein's; the vector modes that hard constraints produce make like defects repel [15], as electric charges do. In three large directions the three together give Newton's $1/r$ potential. A discrete random-geometry model reported to show a Newtonian pull is four-dimensional dynamical triangulations, which did so with test particles moving on a geometry tuned near a transition [19, 20].

V. THE SIMPLEST CANDIDATE, COMPUTED

A graph supports one gapless mode with no further structure: a field ϕ on its points with energy

$$E_\phi = \frac{\kappa}{2} \sum_{\text{links } uv} (\phi_u - \phi_v)^2 = \frac{\kappa}{2}\, \phi^{\mathsf{T}} L\, \phi, \qquad (3)$$

with L the graph Laplacian, whose propagator is the graph's own Green's function and so follows the emergent geometry. The Gaussian integral over ϕ (zero mode removed) adds $\frac{g}{2} \ln \det{}' L$ to the free energy of a wiring, up to a constant the same for every wiring; by the matrix-tree theorem $\det{}' L$ is N times the number τ of spanning trees. If relics are sources q for the field (adding $-q^{\mathsf{T}}\phi$ to E_ϕ), minimizing over ϕ gives a cross term proportional to $-q_1^{\mathsf{T}} L^{+} q_2$ between two relics. We evaluated both on fixed wirings holding two identical defects at every even separation (Table I), each with its expectation written before the calculation. The defect is the curled line of four made by one switch, with a unit charge on each of its points. It is not a relic: with six links it is not a local minimum, and with four it is not the relic of Ref. [2], which adds a square that a single switch from flat space cannot make. For two identical defects, unit charges differ from charges in proportion to energy only by a common factor.

Table I shows fewer spanning trees when the defects are close. Since $\ln \det{}' L$ enters the free energy with a plus sign, under the field reading the fluctuation term then pulls them together (counted instead as hidden structure, an entropy, the trees would push them apart). The

TABLE I. Two identical defects (the curled line of four made by one switch, not a relic) at separation r along an axis, on fixed wirings: the change in $\ln \tau$, τ the number of spanning trees, and in the source cross term $q_1^{\mathsf{T}} L^{+} q_2$ (unit charge on each defect point), each relative to the farthest placement. Exact.

torus	r	$\Delta \ln \tau$	$\Delta\, q_1^{\mathsf{T}} L^{+} q_2$
24×24	2	-0.0478	3.54
	4	-0.0068	1.83
	6	-0.0014	0.91
	8	-0.00034	0.38
$16 \times 16 \times 16$	2	-0.0032	0.369
	4	-0.00017	0.105
	6	-0.000014	0.022

pull is weak and short in reach, falling roughly as r^{-4} in two directions and r^{-6} in three (local exponents 2.8 to 4.9 and 4.2 to 6.2 across the separations of Table I) and worth about $0.024g$ in two directions and $0.0016g$ in three at contact: it passes the test of sign and fails that of shape. The source term, a potential, grows as the relics approach, as the lattice Green's function does, like $1/r$ in three directions, less the images of the periodic torus, and like $-\ln r$ in two. Between a curled torus and flat space, the field's own term favors the curled torus by $0.0144g$ per point with four links, and by $0.0026g$ to $0.0039g$ per point per curled direction with six (one and two curled): the same as lowering λ by about $0.004g$ with four links and $0.001g$ or less with six, too little to move the metastable range of the curled torus near $\lambda = 1.25$.

VI. WHERE THE TWO ROUTES STAND, OCTOBER 2026

The leftovers. The relic is one of three kinds of small knot, read from the wiring of sheets that opened from many seeds [2]: a loop of four points left curled, two neighboring such loops wired like a cube, and a knot of eight points of two kinds. With four links each is a local minimum of its own, and their number is set at the opening and not by the cooling. These are the objects a pull would act on in two directions. With six links none has been found (Sec. II).

The route that needs nothing new: a pull from counting. It needs, first, that what a boundary hides around a leftover grows with the boundary, as in the accounts of Jacobson and Verlinde [5, 6]; an exact count of this kind around a relic has been pre-registered and computed, and awaits its pre-registered reading. Second, that the warm sheet feels both leftovers at once: a fluctuation-induced pull has the range of the surroundings' correlations (Sec. III), so the warm sheet's correlation length is measured first, and a short one closes this route without measuring the pull.

The route that adds something: a quantity on the points. The author has decided to add one on the condition that it matches her account, in which gravity belongs to space, and gives Einstein's numbers wherever they have been measured. The field of Sec. V is the simplest such quantity: one number per point, which, sourced by energy, is Nordström's theory: the right Newtonian limit, no bending of light, and the wrong perihelion shift [18]. So the condition cannot be met by a scalar alone; it needs the rate of time and the curvature of space together.

The nearest published form. In Bianconi's "gravity from entropy" [8] an auxiliary field, the G-field, lives at every point, vanishes in flat empty space and generates a small positive cosmological constant: the closest published relative of a quantity on the points, with equations. Two cautions: its stability is disputed in print (arXiv:2607.18518 and 2609.03082, of which only the abstracts were read), and its discrete form [7], evaluated on this model's own wirings at a uniform metric, rewards squares as Eq. (1) does with $\lambda < 1$, where flat space is not the lowest state [4]. Borrowed whole, it would not keep flat space the ground state. It is a target to compare against, not evidence.

VII. DISCUSSION

What this paper shows is narrow. At fixed wiring and $\lambda > 1$, point defects that share no square do not pull on each other through the energy, and counting symmetries, in the cases computed, gives no attraction that grows as they approach; the gap of flat space argues, by analogy with the published results, that nothing longer-ranged is carried either, whatever the size of the runs. At $\lambda = 1$, which is CQG, we claim only the exact facts. A force with the wiring free to rearrange has not been measured, and with six links the relic it would act on has not been found. The paper does not show that the author's account of gravity is wrong; it argues that this energy lacks the ingredient that the published accounts of a long-range pull need. The simplest ingredient that would supply it, a massless field on the points with relics as its sources, gives Newton's shape, but it puts the pull in by hand; what would be new is that the field travels through the emergent wiring rather than a fixed lattice, and can act back on it. Three tests follow if it is adopted, each to be pre-registered with the author's prediction: whether two relics drift together in a sealed run; the shape of the pull in three directions and its scaling with the relics' energies; and whether many relics clump under it and a dense clump re-curls space, the author's proposed mechanism for black holes [3]. Whether the field could itself arise from the loops of the graph, rather than be added, is the harder question behind all three.

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